SHYAMPUR SIDDHESWARI MAHAVIDYALAYA
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Estd.- 1964

5. (a) Give the recursive definition of pre-order traversal of a binary tree.

(b) Construct a binary tree using the following information (in-order and post-order traversal) :

In-order Traversal : B A D F E G H C

Post-order Traversal : H G F E D C B A

Show all steps.

(c) Write an algorithm to delete an element from a binary search tree which has either one child or has no children. 2+3+5

6. (a) Mention the criteria of binary search technique.

(b) Compare linear and binary search techniques.

(c) Prove that the height of a complete binary tree with n number of nodes is $[\log_2(n+1)]$. 2+3+5

7. (a) What do you understand by collision in hashing?

(b) Discuss any two collision removing techniques.

(c) Explain chaining with an example. 1+(3+3)+3

8. (a) Sort the following elements using Quick Sort technique in ascending order. Show all the steps.

40, 30, 20, 60, 45, 90, 70, 100.

(b) Which sorting technique would you prefer among Quick sort and Merge sort and why? 6+4

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2019

COMPUTER SCIENCE — HONOURS

Paper : CC-3

Full Marks : 50

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question no.*** 1 and ***any four*** questions from the rest.

1. Answer ***any five*** questions : 2×5

(a) Array is a linear data structure. — Justify your answer.

(b) Define De-queue.

(c) Mention two advantages of a threaded binary tree over a binary tree.

(d) Mention one limitation and one advantage of recursion.

(e) Name any four hashing techniques.

(f) Write down any two advantages of circular queue over simple queue.

(g) Compare selection sort and bubble sort with respect to time complexity.

(h) Define binary search tree.

2. (a) Write an algorithm to insert an element in a circular queue.

(b) Convert the following expression into its corresponding postfix expression using stack and evaluate it showing all the steps : 4+(3+3)

$$10 + ((7 - 5) + 10) / 2 + 5*3$$

3. (a) Define min heap.

(b) Construct a min heap with the following elements and then arrange it in ascending order : 200, 10, 90, 60, 100, 50, 150, 40, 20, 70.
Show all the steps. 2+(4+4)

4. (a) Discuss the storage structure of an array.

(b) Represent the following polynomial using array and linked list and then compare them :

$$P(x) = 5x^9 - x^2 + 9x - 50.$$

(c) Write an algorithm to insert an element at the end of the linked list. 3+(1+1+1)+4

Please Turn Over